credential representing a student transcript and the holder shares it with a potential employer, how do they know it's from a real university? There are organizations that accredit universities. These bodies can create trust frameworks for the universities they accredit, and employers can know that the university is real because of that framework.

Despite the advantages decentralized models have, there are several risks and limitations. First, as the university transcript example shows, knowing who to trust is more complicated than in the centralized and federated models. Decentralized systems can be messy. Second, there are adoption challenges. Most internal use cases can be more easily solved using a different model, and external use cases have a multi-sided market problem where issuers are waiting for RPs and RPs are waiting for issuers.

Decentralized identity models can be the means of solving difficult cross-organization authorization problems. In chapter 16 we'll explore a decentralized model that uses verifiable credentials to provide attributes for authorization decisions.

2.6 Trust and confidence

We use the word *trust* a lot in identity and authorization discussions—further evidence of the relational nature of identity. In the chapters that follow, we will distinguish between *trust* and *confidence* to better understand the proper way to solve problems that arise from these important concepts.

2.6.1 The difference between trust and confidence

In a paper published in the *Handbook of Risk Theory* (2012), titled "Risk and Trust," (https://philarchive.org/rec/NICTAR-2), Philip Nickel and Krist Vaesen define trust as the "disposition to willingly rely on another person or entity to perform actions that benefit or protect oneself or one's interests in a given domain." In other words, when we trust someone, we are vulnerable to their actions. There is a risk that they won't follow through or that they're lying.